

Introduction to Modern AI for the Data Domain

From Theory to Application: Mastering LLMs, Vectors, and Agents

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The Enterprise **AI Ecosystem**

How does a modern AI application actually process information and take action?



1. Raw Data

The foundation. This includes your internal Audit Logs, DBs, Route Info, and secure documents.



2. Vector DB

Semantic Storage. Raw data is converted into numerical arrays (embeddings) for rapid context retrieval.



3. LLM Brain

The reasoning engine. Reads the retrieved context to generate accurate, human-readable insights.



4. APIs & Tools

Taking Action. The LLM decides which internal systems to trigger to execute the required task.

Our Environment Today

Zero Budget. Zero Setup. 100% Impact.

Google Colab

Our isolated development environment.

- ▶ Cloud-based Python notebook
- ▶ Runs entirely in your web browser
- ▶ No local installations or dependencies required
- ▶ Perfect for standardizing code across laptops

Google Gemini API

Our reasoning and intelligence engine.

- ▶ Enterprise-grade Large Language Model
- ▶ Generous **Free Tier** for developers
- ▶ No international credit card required
- ▶ Lightning-fast inference speeds

What is an LLM?

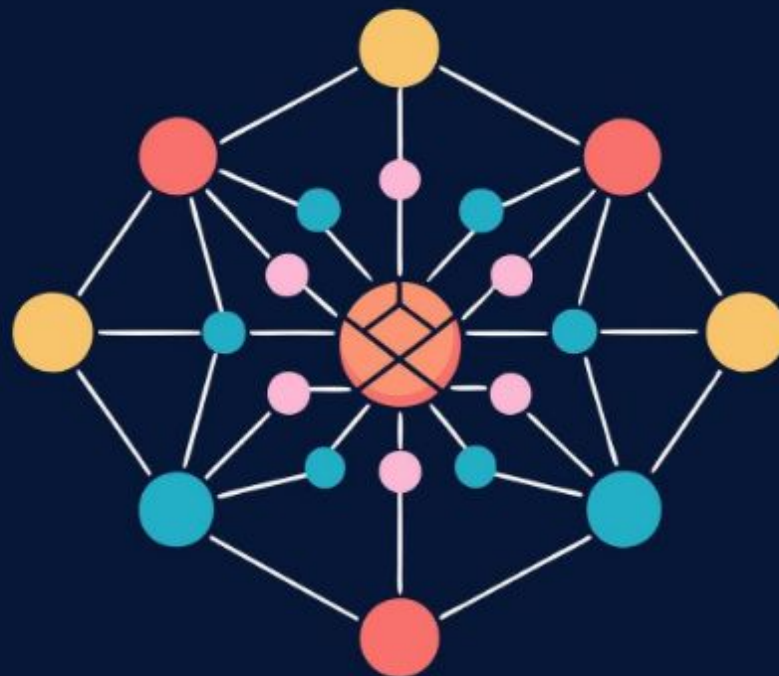
Large Language Models are essentially highly advanced Next-Token Predictors based on massive datasets.

```
> The capital of Sierra Leone is [ Freetown ]
```

The Problem: Hallucinations

If you ask a raw LLM about internal *Transport Authority Server Logs* from yesterday, it will guess (or fabricate information). Why? Because it hasn't been trained on our private enterprise data.

Solution: We must "ground" the model.



Retrieval-Augmented Generation: Giving the AI a custom memory.



1. Document Ingestion Load our private IT Audits, Internal Transport Policies, or proprietary Route Data into the system pipeline.



2. Chunks & Embeddings Break the text into smaller pieces and convert them to numerical vectors so the computer algorithm understands semantic context.



3. Vector Store Save these multidimensional numbers in a specialized database (e.g., ChromaDB) optimized for similarity searches.



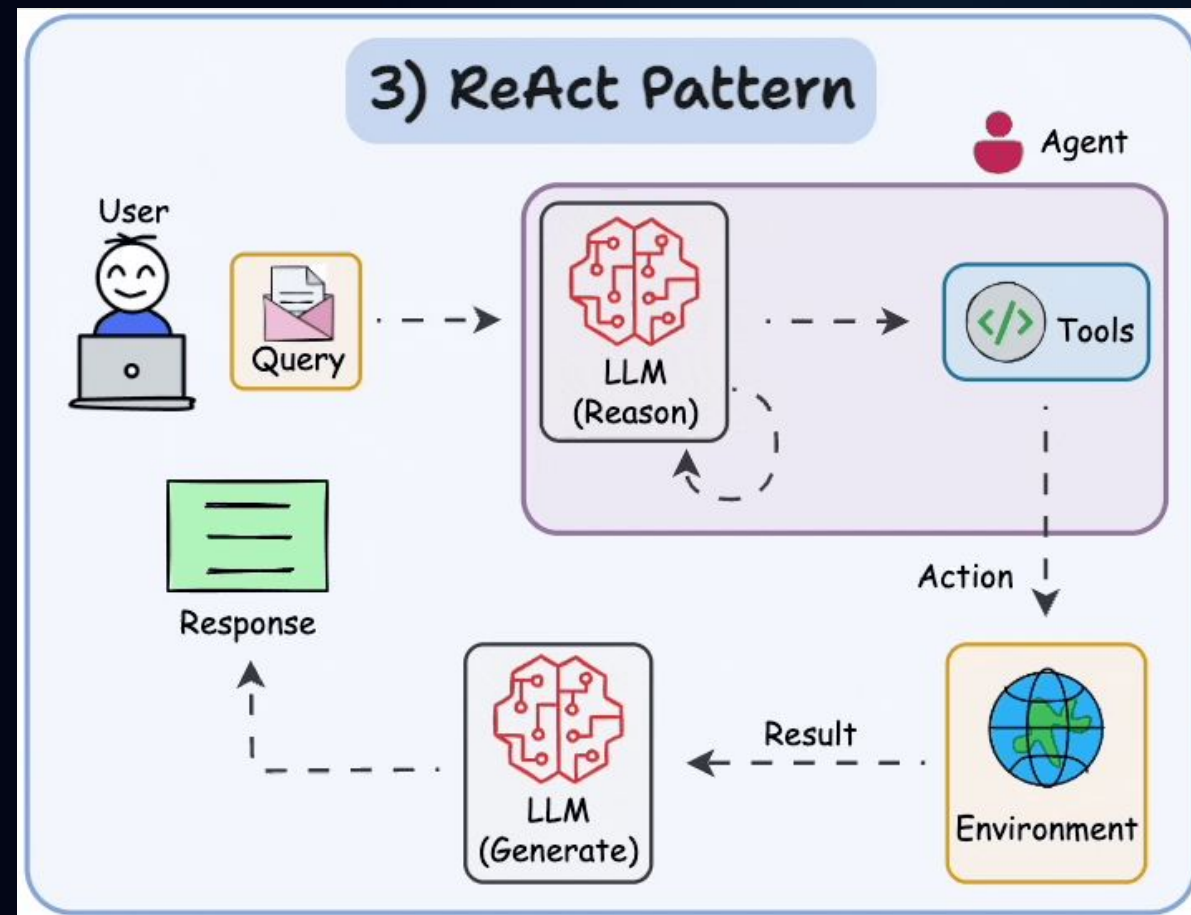
User asks a question → DB retrieves relevant context chunks → LLM reads chunks and answers accurately without hallucinating.

AI Agents (ReAct)

Moving beyond answering questions to **taking independent action**. ReAct stands for Reasoning + Acting.

Goal: "Audit the transit database for yesterday."

- ▶ **Thought:** I need to fetch the logs from yesterday before I can audit them.
- ▶ **Action:** Call Tool `fetch_sql_logs("2026-05-24")`
- ▶ **Observation:** Received 500 rows. One admin downloaded 50GB of data at 3 AM.
- ▶ **Thought:** That is highly anomalous. I will generate an alert.
- ▶ **Final Answer:** "Audit complete. Anomaly detected."





Enough Theory.

Let's Write Some Code.

Please open your Google Colab notebooks.

Questions?

Thank you for your attention.

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